ATHENA TECH

DATA SCIENCE RESOURCES

NLP Projects

Project 1 : Twitter Sentiment Analysis

Problem Statement: Brands, marketers, and political analysts need to understand public opinion in real-time to drive decisions, monitor crises, and adjust campaigns accordingly.

Objective: Develop a sentiment analysis model that classifies tweets as positive, negative, or neutral using supervised learning or transformer-based models.

Dataset Sources

- Sentiment140 Dataset
- Twitter US Airline Sentiment Dataset

Techniques & Concepts

- Text preprocessing (cleaning, tokenization, lemmatization)
- TF-IDF or word embeddings (Word2Vec, GloVe)
- Machine learning (Logistic Regression, Naive Bayes) or deep learning (LSTM, BERT)
- Visualization (word clouds, sentiment trends)

Tools & Libraries: Python, NLTK, Scikit-learn, Keras, TensorFlow, Hugging Face Transformers, Tweepy

Expected Outcome: A web or script-based application that takes a tweet or hashtag and returns sentiment classification with accuracy metrics.

Project 2: Resume Parser Using NLP

Problem Statement: Hiring teams need to extract structured information like name, skills, experience, and education from thousands of unstructured resumes quickly and accurately.

Objective: Create an automated parser that uses Named Entity Recognition (NER) and regular expressions to extract key details from resumes in PDF or DOCX formats.

Dataset Sources

- Resume Dataset (Kaggle)
- Custom collected resume samples in .docx or .pdf format

Techniques & Concepts

- PDF and DOCX parsing
- Named Entity Recognition (NER)
- Regex-based keyword extraction
- Skill matching using dictionaries or embeddings

Tools & Libraries: Python, spaCy, PyPDF2, docx2txt, NLTK, pandas

Expected Outcome: A tool or API that extracts structured resume data and outputs it in JSON or CSV format, ready for ATS integration.

Project 3: News Article Classification

Problem Statement: News platforms and aggregators need to categorize large volumes of articles into topics like politics, business, and technology to enhance navigation and personalization.

Objective: Train a multi-class text classification model to predict the category of a news article based on its headline and content.

Dataset Sources

- AG News Dataset
- News Category Dataset

Techniques & Concepts

- TF-IDF and CountVectorizer
- Naive Bayes or Multinomial Logistic Regression
- BERT or RoBERTa for advanced contextual classification
- Cross-validation and evaluation (F1-score, confusion matrix)

Tools & Libraries: Python, Scikit-learn, Hugging Face Transformers, TensorFlow, pandas

Expected Outcome: A script or application that classifies a new article into categories like world, tech, sports, or health.

Project 4: Text Summarization with Transformers

Problem Statement: Professionals need to read lengthy documents, news articles, or reports. Automated summarization can save hours and improve productivity.

Objective: Build a text summarization model that generates concise summaries of longer texts using extractive or abstractive techniques.

Dataset Sources

- CNN/DailyMail Summarization Dataset
- Newsroom Dataset

Techniques & Concepts

- Extractive methods (TextRank, LexRank)
- Abstractive summarization with transformers (BART, T5, Pegasus)
- Tokenization and input chunking
- ROUGE scoring for summary evaluation

Tools & Libraries: Python, Hugging Face Transformers, TensorFlow, spaCy, NLTK

Expected Outcome: A functional summarizer that reduces long-form articles into readable 3–5 sentence summaries suitable for newsletters, dashboards, or reports.
